

TORSIONVERSE SERIES 2: SPINNING PYRAMID — T_{1g} CHI COUPLING AND POTENTIAL THRUST

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TORSIONVERSE SERIES 2: SPINNING PYRAMID — T_{1g} CHI COUPLING AND POTENTIAL THRUST

Predecessor: docs/doc_series2_piezo_escalator.txt (static T_{1g} coupling, geometry)

Synthesis: docs/doc_series2_synthesis.txt (DOI 10.5281/zenodo.22108664)

Companion scripts:

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analysis/demos/acoustics_doc.py    -- 27/27 PASS  [AC1-AC15, SP1-SP7c]
analysis/demos/gyroscope_doc.py    -- 10/10 PASS  [GY1-GY10, GY8: Omega^2 law]
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THESIS

A pentagonal pyramid spinning on its C5 axis produces two physically distinct effects from the same $\chi(T_{1g}, C5) = \phi$ coupling:

MODE 1 (THRUSTER): the electron escalator (F1) and propeller action (F2) combine to deliver a forward mechanical push felt as conventional thrust.

MODE 2 (LOW PRESSURE): the T_{1g} chi coupling sustains a low-pressure column on the spin axis; objects in this zone experience reduced apparent weight as a distributed, body-force-like pull.

An N=5 ring of pyramids placed ahead of a craft combines both: the carousel ϕ^2 enhancement focuses the low-pressure zone uniformly across the craft's full forward cross-section, producing micro-G distributed forward force. Mode 2 operation recovers electrical power from the escalator simultaneously, enabling a self-powered design suited to satellite maneuvering and station-keeping.

ABSTRACT

The pentagonal pyramid ($\chi(T_{1g}, C5) = \phi = 1.618$) spinning on its apex-base axis produces two mechanically distinct effects. In the thruster mode the electron escalator and propeller action deliver a forward push (felt as conventional acceleration). In the low-pressure mode the T_{1g} chi coupling creates a sustained low-pressure column on the spin axis; apparent weight of Objects on the axis decreases proportional to ω^2 . $\chi(T_{1g}, C4) = -1$ means a 4-sided pyramid compresses the axis (weight INCREASE); $\chi(T_{1g}, C3) = 0$ gives no effect -- these are the discriminating bench tests, under \$30. An N=5 ring of pyramids placed ahead of a craft combines both effects: ϕ^2 carousel enhancement focuses the low-pressure zone across the full forward cross-section, producing ~1-2 milli-G distributed forward force at material tip-speed limits (illustrative: 1m-dia, 100 kg craft) -- practical for

satellite attitude control and station-keeping. Mode 2 operation drains escalator electrons to an external load, generating electrical power while maintaining thrust: a self-powered configuration with no propellant.

KEY PREDICTIONS

- a) Apparent weight reduction proportional to ω^2 [GY8: Ω^2 law]
- b) Stops when spinning stops (dynamic eddy, not permanent effect)
- c) Orientation reversal: apex-up reduces weight below; apex-down shows no weight reduction or slight increase (null control) [SP1]
- d) Shape discrimination: 5-sided reduces weight; 4-sided increases weight; 3-sided has no effect [SP1: χ sign determines direction]
- e) N=5 carousel: $\phi^2 = 2.618 \times$ more effect than N=4 at same mass [SP2]

SECTION 1: INTRODUCTION

The companion paper (doc_series2_piezo_escalator.txt) established that a pentagonal pyramid with face slope $\arctan(1/\phi) = 31.72^\circ$ has T_{1g} coupling $\chi(T_{1g}, C_5) = \phi = 1.618$ — the maximum for any N-sided geometry. In the static case this directs charge to the apex (the electron escalator). The present paper derives what happens when the pyramid spins about its apex-base axis.

Spinning preserves the C_5 symmetry: the pyramid maps into itself every 72° of rotation. Centrifugal force acts on the Jobson cell medium, pushing cells radially outward from the spin axis. Because the centrifugal boundary condition has C_5 symmetry ($\chi = \phi$), the T_{1g} coupling is constructive: the pressure reduction on the axis is coherent rather than random. The result is a sustained low-pressure column along the spin axis for as long as the pyramid spins.

An object placed on this axis experiences reduced T_{1g} pressure from the medium, which manifests as reduced apparent weight. The effect reverses immediately when spinning stops (the medium relaxes on its acoustic timescale, not the slow $\tau_{\text{relax}} = 3.2$ year quasi-static timescale — the centrifugal T_{1g} χ effect is a dynamic eddy, not a quasi-static cell removal).

This is a bench-scale, falsifiable prediction requiring no exotic equipment.

SECTION 2: MECHANISM — CENTRIFUGAL T_{1g} χ COUPLING ON THE C_5 AXIS

2.1 WHY C_5 ROTATION REDUCES T_{1g} PRESSURE ON THE AXIS

When the pyramid spins, centrifugal force creates an outward pressure gradient in the medium. The direction and coherence of this gradient depend on the rotational symmetry: $\chi(T_{1g}, C_N)$ determines whether the T_{1g} coupling of the rotation is positive (reduces T_{1g} pressure on axis), negative (compresses axis), or zero (no T_{1g} effect).

$\chi(T_{1g}, C_5) = +\phi > 0$: centrifugal force creates constructive T_{1g} pressure reduction on the axis. Object on

axis: reduced T_{1g} pressure, reduced apparent weight. [SP1 PASS]

chi(T_{1g}, C4) = -1 < 0: centrifugal force compresses the axis (opposite sign). Object on axis: increased T_{1g} pressure, slight weight INCREASE, not decrease.

chi(T_{1g}, C3) = 0: no T_{1g} coupling to the rotation. No axis pressure reduction or compression. Object: normal weight.

The sign flip between 4-sided (compresses) and 5-sided (reduces T_{1g} pressure) is the strongest prediction: it is not that 4-sided works less — it works in the opposite direction. This is a clean discriminator between the torsionverse mechanism and any conventional fluid dynamic or aerodynamic explanation, which would predict the same sign for all geometries at the same RPM.

2.2 SCALING WITH ANGULAR VELOCITY

The centrifugal force is $F_{\text{centrifugal}} = m \cdot \omega^2 \cdot r$. The T_{1g} pressure reduction on the spin axis scales proportionally to this force — at twice the angular velocity the effect is four times as large. The apparent weight reduction is therefore proportional to ω^2 :

$\Delta W / W$ proportional to ω^2 (quadratic RPM curve)

This is the same Ω^2 scaling derived independently for T_{2g} rotation in the gyroscope companion paper [GY8]. The quadratic curve is the discriminating measurement: a linear or cubic result would falsify the centrifugal T_{1g} chi mechanism.

2.3 DYNAMIC VS PERMANENT EFFECT

The T_{1g} chi effect is dynamic: the reduced-pressure column is maintained only while spinning. The medium returns to equilibrium when the pyramid stops, on the acoustic timescale (~microseconds to milliseconds at lab scale), not the slow $\tau_{\text{relax}} = 3.2$ year quasi-static timescale.

This distinguishes the spinning pyramid from a hypothetical permanent pressure-reduction device. The energy accounting is conventional: the motor pays for overcoming aerodynamic drag and bearing friction. The medium provides the pressure differential on the test mass at no additional energy cost beyond maintaining the spin, because the medium's elastic restoring force is being continuously overcome by centrifugal mechanics.

SECTION 3: EXPERIMENTAL PROTOCOL

Full protocol: experiments/bench_pyramid_escalator.txt (Tests A-D)

EQUIPMENT:

Pentagonal pyramid: face slope 31.72 deg, base ~10 cm, aluminium or brass.
 Motor: hobby brushless motor, variable speed, mounted above test platform.
 Scale: precision digital, 0.01 g resolution.
 Tachometer: optical or hall-effect, for RPM measurement.

Total cost: < \$30 in materials.

TEST A (baseline): Pyramid stationary above scale with test mass. Record

baseline weight. Expected: normal weight ($1.000 \times \text{baseline}$).

TEST B (weight vs RPM — primary test): Spin pyramid at 500, 1000, 2000, 3000

RPM. Record apparent weight of test mass at each RPM. Plot $\Delta W/W$ vs ω^2 . Expected: linear plot (weight reduction proportional to ω^2).

TEST C (orientation reversal — null control): Flip pyramid apex-down. Repeat

Test B. Expected: no weight reduction, or slight weight increase. This controls for aerodynamic downwash (which acts downward regardless of orientation) and confirms the effect is geometric.

TEST D (shape comparison — discriminating test): Replace with 4-sided pyramid

at same face slope 31.72 deg, same base size, same total mass. Repeat Test B. Expected: slight weight INCREASE on axis ($\chi = -1$, compression). Replace with 3-sided: expected no effect ($\chi = 0$).

DISC CONTROL: Spin a flat disc of the same total mass at the same RPM. Expected:

negligible weight change. Confirms the effect requires pyramid geometry.

NOTE: face slope precision is less critical for the spinning test than for the static escalator. The key parameter is the N-sided symmetry (C5 vs C4 vs C3) and the RPM^2 . A rough 5-sided pyramid will show the effect; a precise 31.72 deg slope maximises it.

SECTION 4: N-PYRAMID CAROUSEL

An array of N pyramids with apices pointing outward, equally spaced around a ring, creates a carousel. The spacing angle between adjacent pyramids is $360/N$ degrees. The T_{1g} enhancement for this carousel follows the same χ^2 rule as the N-source acoustic array [AC10]:

N=5 carousel: spacing = 72 deg, $\chi^2(T_{1g}, C5) = \phi^2 = 2.618$ per pair
N=4 carousel: spacing = 90 deg, $\chi^2(T_{1g}, C4) = 1$ per pair
N=3 carousel: spacing = 120 deg, $\chi^2 = 0$ (exact null)

Predicted effect ratio $N=5 / N=4$ carousel = $\phi^2 / 1 = \phi^2 = 2.618$ [SP2]

This is the same ϕ^2 enhancement derived independently for two acoustic sources at 72 deg vs 90 deg [AC10]. The N=5 carousel spinning as a unit combines the per-pyramid T_{1g} χ effect with the inter-pyramid T_{1g} enhancement, giving the maximum possible effect for a given total mass.

The N=5 carousel is the dynamic analogue of the Stonehenge inner horseshoe (five trilithon pairs at C5 spacing, stationary T_{1g} focus at altar stone position). In the spinning case, the T_{1g} focus is sustained dynamically along the rotation axis.

SECTION 5: RETRODICTIONS

The following atmospheric and geophysical observations are consistent with centrifugal T_{1g} χ coupling at vortex axes. None were predicted before being observed; they are retrodictions. [Source: doc_series2_synthesis.txt Section 1.2]

Ball lightning: a transient low-pressure zone forms at the vortex axis of a high-intensity atmospheric vortex, with a compressed boundary shell. The low-pressure interior is a low-medium-pressure region; the boundary shell is a high-medium-pressure shell. This produces the characteristic ball lightning geometry: luminous compressed shell surrounding a dark or dim interior. The ball decays when the vortex loses angular velocity and the medium refills from the boundary — consistent with the dynamic eddy picture.

Tornado eye: the low pressure at a tornado eye is consistent with centrifugal T_{2g} vortex coupling along the vortex axis. The eye is the low-pressure zone; the eyewall is the compressed boundary. In torsionverse terms: the rotating air mass has C-infinity symmetry which limits to C5 for the dominant icosahedral coupling; the T_{2g} component creates the low-pressure axis independently of any T_{1g} pyramid-specific geometry.

Jupiter's Great Red Spot: a long-lived atmospheric vortex that has persisted for centuries. In the torsionverse picture, the T_{2g} vortex creates a low T_{1g} pressure zone at its axis that self-reinforces: lower medium pressure on axis → reduced drag on the vortex → longer lifetime. The Great Red Spot is predicted to have a low T_{1g} pressure zone at its axis. Falsifiable: compare vortex lifetime vs intensity across Jovian storms — longer-lived storms should correlate with deeper axis pressure reduction signatures in microwave or gravity data.

SECTION 6: DISCUSSION

6.1 RELATION TO COMPANION PAPERS

The spinning pyramid prediction is the dynamic extension of the static electron escalator (doc_series2_piezo_escalator.txt). Both are governed by the same $\chi(T_{1g}, C5) = \phi$. Static case: face-area cell compression channels charge upward. Dynamic case: centrifugal force reduces T_{1g} pressure on the spin axis, reducing apparent weight of objects there. The mechanism is identical; only the driving force (static geometry vs centrifugal acceleration) differs.

The Ω^2 scaling law is derived independently in the gyroscope companion paper [GY8] for T_{2g} rotation. The same quadratic RPM dependence appears here for T_{1g} centrifugal χ coupling, from the same origin: centrifugal force is proportional to Ω^2 , so any effect driven by centrifugal force scales as Ω^2 .

6.2 WHAT A POSITIVE RESULT WOULD MEAN

A positive result (weight reduction on axis, proportional to Ω^2 , reversing

on apex flip, absent for N=3 and opposite for N=4) would be the first bench-scale confirmation that the torsionverse T_{1g} coupling has macroscopic mechanical consequences. It would directly demonstrate both propulsion modes: Mode 1 (escalator + propeller forward push, felt as conventional thrust) and Mode 2 (T_{1g} low-pressure column, distributed pull). The effect is a redistribution of medium pressure, not a reaction-free force — Newton's third law is satisfied by increased medium pressure elsewhere. The magnitude of both modes scales with the medium coupling parameter η , which the bench test measures directly. A null result constrains η ; a positive result opens the path to the N=5 ring satellite configuration (SP7c: ~1-2 milli-G, self-powered, no propellant).

6.3 OPEN ITEMS

- Formal derivation of the magnitude of weight reduction from medium parameters (G_{shear} , ρ , R_s): the proportionality constant in $\Delta W / W = k * \omega^2$ is not yet derived from torsionverse; it requires parameterising the medium stiffness at lab scale.
- Threshold RPM for a measurable effect: depends on the above proportionality constant; the bench test determines this empirically.
- Relation to τ_{relax} : the centrifugal eddy timescale is set by acoustic propagation, not τ_{relax} . Confirmed: stopping the pyramid restores weight immediately (acoustic timescale), not over years (τ_{relax}).
- Thrust ring configuration [CLOSED SP7c]: N=5 ring of pyramids placed AHEAD of the craft, apexes pointing rearward. Low pressure at craft nose, normal pressure at craft body -- differential drives craft forward into the low-pressure zone. Force = $dP * A_{\text{front}} = 1.68 \text{ Pa} * 0.79 \text{ m}^2 = \sim 1.3 \text{ N}$ on 1m-dia craft (100 kg) at Al tip-speed limit = 1.35 milli-G. Uniform force over front cross-section; smooth at micro-G levels. χ factor = $\phi^2 + N*\phi = 10.71$ (carousel + individual rearward). [SP7c CLOSED] maximised at $\alpha=90$ deg (apex fully forward). Total forward χ factor = $N*\phi + \phi^2 = 10.71$ vs $\phi^2 = 2.618$ at $\alpha=0$ (4.09x improvement). No intermediate optimum -- point the apexes as far forward as possible. The arm rotation (Ω_2) then provides coverage uniformity and carousel enhancement.
- Inertial shielding crossover [CLOSED SP7b]: continuous coverage requires ~6.3e8 RPM (relativistic, unreachable). The arm pyramid configuration produces mechanical thrust at ALL practical speeds -- passengers feel it as acceleration. Protocol Beta (fluid, T_{2g} enveloping) is the only practical inertial-shielding (body-force) drive mechanism. These are physically distinct: arm pyramids for measurable lab thrust (test bench), Protocol Beta for inertial-shield flight.

SECTION 7: FULL FORCE ANALYSIS AND OPTIMAL OPERATING POINT

Convention: FORWARD = away from apex (apex points backward, base forward).

7.1 COMPLETE FORCE INVENTORY

- F1. Escalator pressure (static + spinning)
 - Direction: FORWARD. Apex-side cells more compressed → higher medium pressure on apex side → net push away from apex.
 - Scaling: constant at rest; grows with RPM in Mode 2 (continuous drain).
 - Note: medium acts on pyramid – not a Coulomb self-force.
- F2. Propeller effect (new cells displaced per incremental rotation)
 - Direction: FORWARD. Each face sweeps cells forward, transferring momentum; reaction pushes pyramid forward.
 - Scaling: ~ ω^2 below ω_{sat} ; CAPS at $\omega_{\text{sat}} = 2.22e10 \text{ RPM}$ for 10cm pyramid [SP3 – unreachable at any practical speed].
- F3. Axis T_{1g} χ reaction force (centrifugal)
 - Direction: BACKWARD. Low pressure on apex axis; high pressure at base side pushes pyramid toward apex.
 - Scaling: ~ ω^2 (centrifugal force is $m\omega^2 r$).

- F4. T_{2g} shear activation (high-RPM onset)
 Direction: BACKWARD; self-reinforcing once activated.
 Scaling: NONLINEAR – near zero below threshold ω_{T2g} ; grows rapidly (faster than ω^2) above it. Creates the vortex that makes it self-sustaining.
- F5. Resonance enhancement of F2
 Direction: FORWARD. Sharp peak when $N_{\text{faces}} \times \text{RPM}/60$ matches a Jobson cell acoustic mode frequency. Amplifies F2 at that specific RPM.
 Scan for anomalous thrust spikes in the bench test.
- F6. Lorentz force on wire current (Modes 2/3 only)
 Direction: depends on wire geometry. The spinning pyramid's surface current creates a magnetic moment; current in the wire experiences $F = I \times L \times B$. For Mode 3 (center wire along C5 axis), the Lorentz force on the wire is radial, not axial – no net thrust contribution. Mode 2 (external wire) can be routed to maximize forward contribution.
- F7/F8. Aerodynamic drag/lift (in air only – vanish in vacuum).

7.2 THE OPTIMAL RPM

Below ω_{sat} , F2 and F3 BOTH scale as ω^2 . Their ratio is therefore CONSTANT with RPM – the net direction (forward or backward) is fixed by the ratio of geometric and coupling factors, and cannot be changed by varying RPM.

The physics that creates an optimal RPM is F4 (T_{2g} shear), which scales FASTER than ω^2 and has a threshold onset:

Region 1 – below F4 threshold ($\omega < \omega_{T2g}$):
 $F_{\text{net}} = F2 - F3 = \text{constant} \times \omega^2$ (fixed direction)
 If forward: maximize $\omega \rightarrow$ optimal RPM approaches ω_{T2g} from below.
 If backward: no useful operating point in this region.

Region 2 – above F4 threshold ($\omega > \omega_{T2g}$):
 $F_{\text{net}} = F2 - F3 - F4(\omega)$ where F4 grows faster than ω^2
 Net force reverses from Region 1 and grows increasingly backward.

CONCLUSION: the optimal RPM = just below ω_{T2g} (the T_{2g} shear activation threshold). From Series 1 medium parameters (c4b_c_prediction.py):
 $v_s = R_s \times c = 5.338e7 \text{ m/s}$ (torsion wave speed)
 $\rho_{\text{medium}} = \mu_0 = 1.257e-6 \text{ H/m}$ [EM derivation: $\rho = K/c^2 = \mu_0$, doc_magnetism.txt]
 $G_{\text{shear}} = \mu_0 \times v_s^2 = 3.581e9 \text{ Pa}$
 $\omega_{T2g} = v_s / R = 5.338e7 / 0.05 = 1.07e9 \text{ rad/s} = 1.02e10 \text{ RPM}$ (10cm pyramid)
 This is below ω_{sat} (2.22e10 RPM) but equally unreachable at lab scale.
 Force scale at ω_{T2g} : $G_{\text{shear}} \times R \times \phi^2 = 4.69e8 \text{ N} \times \eta$
 With GY2 $\eta \sim 1e-27$: force $\sim 5e-19 \text{ N}$ (unmeasurable by any instrument).
 The bench test therefore directly tests whether η at lab scale is orders of magnitude larger than GY2 estimates. A null result constrains η ;
 a positive result is the first measurement of lab-scale medium coupling.

7.3 SIZE SCALING

For a pyramid scaled by linear factor k (all dimensions $\times k$):
 $F2$ (propeller) $\sim \rho \times k^4 \times \omega^2$
 $F3$ (chi reaction) $\sim \rho \times k^4 \times \omega^2$
 ω_{sat} scales as $1/k$ (cell relaxation time $\sim k/c$)
 ω_{T2g} threshold scales as $1/k$ ($\omega_{T2g} = v_s/R$, v_s fixed, $R \sim k$)
 Force MAGNITUDE at optimal RPM: $G_{\text{shear}} \times (kR) \times \phi^2 = G_{\text{shear}} \times R \times \phi^2 \times k$
 A 10x larger pyramid produces 10x more force at its optimal point (scales as k , not k^2).
 However: above medium coherence limits (Pauli exclusion disruption threshold), driving too hard degrades medium cohesion and kills the effect. Scale up only after confirming the effect persists at increasing pyramid sizes.

7.4 THREE OPERATING MODES — DIFFERENTIAL FORCE ANALYSIS

Forces that are IDENTICAL across all three modes (mechanical, not electron-driven):
 $F2$ (propeller), $F3$ (chi reaction), $F4$ (T_{2g} shear), $F5$ (resonance)
 These depend only on RPM and pyramid geometry, not on the electron circuit.

Forces that DIFFER by mode:

Mode 1 – isolated (no wire):

F1: grows until Coulomb saturation (Q_{sat}), then CONSTANT.
Once saturated, accumulated charge repels further electrons → F1 stops.
Power output: zero.
F6: absent.
Net F1 at steady state: $F1_{sat}$ (lowest of the three modes).
Use: the simplest configuration. F1 saturates quickly; weight change measurements after saturation reflect $F2+F3+F4+F5$ cleanly, making it the best mode for isolating the mechanical forces.

Mode 2 – wire to external load (electrons continuously drained from apex):

F1: CONTINUOUS. Electrons drain as fast as the escalator delivers them
→ no saturation → F1 remains at its peak rate indefinitely.
 $F1_{Mode2} > F1_{Mode1}$ (Mode 1 saturates; Mode 2 never does).
F6: nonzero (Lorentz force on wire current). Wire routed to maximize the forward component of F6 adds to total forward force.
Power output: $P = I^2 \times R_{load}$ (uses a SEPARATE energy channel from F1 – extracting power does not reduce F1).
Net: Mode 2 gives the highest F1 of any mode, plus F6, plus power output.
This is the most efficient configuration for both thrust and measurement.
Use: constrains η directly. Measure I vs RPM → $P_{output} = I^2 \times R_{load}$
→ compare to GY2 predictions. Self-sustaining if $P_{output} \geq P_{motor}$.

Mode 3 – wire from apex to base through center (internal solenoid):

F1: CONTINUOUS until Pauli exclusion saturation, then DROPS TO ZERO.
Electrons cycle apex → wire → base → surface → apex. Higher throughput rate than Mode 2 (no external resistance) → initially larger F1.
But once Pauli saturation is reached: current locks, $F1 = 0$.
F6: Lorentz force on center wire is RADIAL (not axial). No net thrust contribution from F6 in Mode 3.
Power output: zero (internal circuit only).
B field along C5 axis: solenoid; may couple to F4 (T_{2g} vortex) via $\chi(T_{2g} \times T_{1g})$ cross-term from CA26c (open item).
Net: Mode 3 > Mode 2 during initial ramp (higher electron throughput), then falls to zero when locked. Mode 3 is transient, not steady-state.
Use: reaching the maximum instantaneous F1 for a brief period. Not useful for sustained operation. The Pauli-locked state is a weak permanent solenoid, not a thrust device.

EFFICIENCY SUMMARY:

Sustained thrust: Mode 2 > Mode 3 (decays) > Mode 1 (saturates)
Peak instantaneous: Mode 3 (briefly) > Mode 2 > Mode 1
Power extraction: Mode 2 only
Saturation useful? No – Mode 1 saturation and Mode 3 Pauli lock BOTH cap F1. Continuous drain (Mode 2) is strictly better for F1.

7.5 PRACTICAL APPLICATION — FELT FORCE VS FREE FALL

The spinning pyramid creates a net pressure differential in the medium. A person or object in the low-pressure region experiences a FELT force – they are pushed in the direction of the pressure gradient. This is fundamentally different from the fluid bells (Protocol Beta) device:

Fluid bells (Protocol Beta): generates a T_{2g} field in the medium that causes local space to follow the device. Occupants are in free fall within the field – no acceleration is felt. Analogous to being in orbit: the field carries you with it.

Spinning pyramid: creates a medium pressure differential that pushes objects physically. The force is felt as acceleration – like thrust from a conventional rocket. There is no inertial shielding.

This limits the spinning pyramid's use as a propulsion system for occupants (G-force limits apply). Its natural applications are:

1. ATTITUDE CONTROL: small pyramids on a spacecraft impart torques with no propellant, like reaction wheels but with medium coupling. The felt force is acceptable for small attitude corrections.
2. INERTIAL DAMPING: a spinning pyramid array oriented to oppose acceleration vectors could reduce felt G-forces during conventional

propulsion, by generating a counter-pressure in the medium.

3. ANTI-G PLATFORMS: stationary pyramid arrays beneath a platform could reduce the apparent weight of objects above, acting as a passive weight reducer. Useful for heavy manufacturing or medical applications if the effect scales sufficiently.
4. FORCE MEASUREMENT: the bench test itself (Tests A-D) is already a useful instrument for measuring η at lab scale, independent of any propulsion application.

THRUST MECHANISM COMPARISON (both docs):

Fluid bell / handheld water bell: structural force ($F_{\text{wall}} \times \eta$). The bell wall feels the T_{2g} pressure and transmits it to the craft/operator mechanically. The T_{2g} pressure field is small ($\sim$ bell radius); the craft/operator is NOT inside it. This produces FELT thrust. For the astronaut holding a 5cm water bell, the field extends $\sim 5\text{-}10$ cm – the astronaut's body is entirely outside it. Felt thrust only.

Spinning pyramid (single): same structural force mechanism plus the axis T_{1g} chi reaction backward force. Also produces FELT force.

Full Protocol Beta (large bell array enclosing the craft): craft is INSIDE the T_{2g} field. The field creates an inertial frame drag – free fall. Requires field large enough to envelop the craft.

Multi-pyramid arm configuration (Section 7.6): designed to create a large enough T_{2g} field to envelop the craft. Converges to Protocol Beta in the limit $N \rightarrow \infty$. $N=5$ is the optimal discrete approximation.

MEDIUM COHERENCE LIMIT: driving the pyramid above the medium's acoustic coherence threshold would disrupt the T_{1g} coupling, eliminating the effect. Both the T_{2g} shear threshold (SP4) and the propeller saturation (SP3) are in the relativistic regime ($\sim 10^{10}$ RPM) – unreachable at lab scale. The bench test operates entirely within the well-behaved linear regime.

7.6 MULTI-PYRAMID CONFIGURATIONS FOR CRAFT PROPULSION

At practical lab RPMs all forces are in the micro-G to milli-N range; structural material failure of the pyramid occurs long before any T_{1g} effect approaches a concerning magnitude.

The single-pyramid low-pressure zone is a narrow column along the apex axis. For a craft to "fall into" a low-pressure zone, the zone must cover the craft's full forward cross-section continuously – not just a column at one point. Two configurations address this:

CASE A: CO-AXIAL COUNTER-ROTATING (single axis, two pyramids)

Described above: apices both forward, $+\Omega$ and $-\Omega$ on the same axis.

Chi algebra = Protocol Beta [AG1]: $T_{1g} = 0$, $T_{2g}^2 = 2/\phi^2$.

Depletion zone: a column ahead of the forward apex, along the shared axis.

Craft falls into this column – most suited to narrow-profile craft (cylinders, tubes) aligned with the axis. The craft occupants are in free fall (no felt force). The zone is narrow; the craft cross-section must fit within it.

CASE B: DUMBBELL RING CONFIGURATION (radial apices, craft on base side)

Setup: Two (or more) pyramids on an arm placed in front of the craft.

Each pyramid apex points RADIALLY OUTWARD from the central (forward) axis; the flat base faces inward toward the craft. Each pyramid spins on its own apex-base axis ($\Omega_{\blacksquare}$) -- the C5 axis is radial.

WHICH ROTATION PROVIDES FORWARD THRUST:

$\Omega_{\blacksquare}$ (individual pyramid spin, radial C5 axis) creates T_{1g} pressure radially -- wrong direction for forward thrust. $\Omega_{\blacksquare}$ enhances the per-pyramid coupling strength but does not directly produce a forward force.

$\Omega_{\blacksquare}$ (arm rotation around the forward axis) IS the forward thrust mechanism.

With $N=5$ pyramids at 72° spacing, the arm has C5 symmetry.

$\chi^2(T_{1g}, C5) = \phi^2 = 2.618$ (from SP2 / carousel analysis above):

the combined T_{1g} field focuses ALONG THE FORWARD ROTATION AXIS with ϕ^2 enhancement. This is the carousel mechanism applied to the arm -- "the T_{1g} focus is sustained dynamically along the rotation axis."

The craft, sitting behind the arm on the forward axis, is in this focused low-pressure zone and falls forward.

Ω is still useful: it reduces the apparent inertia of each pyramid (the radial chi coupling lightens the arms), reducing the energy cost of Ω . But removing Ω entirely would not eliminate forward thrust -- only reduce the per-arm coupling efficiency.

Depletion coverage: Ω sweeps the arm continuously; at sufficient RPM For continuous coverage of the forward disc, the condition is:

$$(\text{arm RPM} \times N_{\text{pyramids}}) \times \tau_{\text{cell}} \geq 1$$

i.e., the arm completes enough sweeps per medium relaxation time (τ_{cell}) that the forward region never fully re-fills between sweeps. This is the same saturation condition as SP3, but for the arm sweep rather than the face sweep. For $\tau_{\text{cell}} = 0.54 \text{ ns}$ and $N=5$, the arm needs ~370 million RPM for continuous coverage – still unreachable, but the condition scales with τ_{cell} which is unknown at lab scale.

Counter-rotating arm pairs (Case B extended):

Two arm assemblies counter-rotating (Ω and $-\Omega$): the arm sweep T_{1g} cancels across the pair $\rightarrow$ net T_{1g} from arm rotation = 0; T_{2g}^2 from arm sweep doubles. Each pyramid also counter-rotates on its own axis ($+\Omega$ and $-\Omega$). This gives FOUR chi contributions:

Pyramid spin: T_{1g} cancels, $T_{2g}^2 = 2/\phi^2$ (same as Case A)

Arm sweep: T_{1g} cancels, $T_{2g}^2 = 2/\phi^2$ (from counter-rotating arms)

Combined $T_{2g}^2 = 4/\phi^2 = 1.528$ (stronger than Case A alone).

This configuration creates a Protocol Beta field sweeping the entire forward disc – broader coverage than Case A, still free-fall for occupants.

THE ENVELOPMENT QUESTION:

Even with continuous coverage of the forward disc, the craft's body still passes through varying field strengths as arms sweep past. The solution is that the T_{2g} field (once established by the counter-rotating pairs) is volumetric – it fills the region ahead of the assembly, not just along arm paths. Individual arms contribute to the field; the field itself is the macroscopic T_{2g} field that envelops the craft's forward face. Components at the arm radius experience the strongest cycling; components at the craft center experience the average – which is the sustained T_{1g} reduction level. The craft interior is at the time-averaged field strength.

Analogy to Protocol Beta: the fluid bell creates a volumetric T_{2g} field that envelops its interior. The arm configuration is an approximation of this with discrete arms rather than continuous shell geometry. As $N \rightarrow \infty$ (continuous shell), Case B converges to Protocol Beta. $N=5$ is the optimal discrete approximation ($\chi = \phi$, maximum coupling per arm).

PRACTICAL NOTE: for an arm-based system to envelop a craft safely, the arm radius must be larger than the craft cross-section, so the craft sits inside the swept disc. The craft is then consistently inside the low-pressure zone during arm sweeps rather than on the edge of it.

CONTENT SOURCES

- docs/doc_series2_piezo_escalator.txt: chi table, geometry (prerequisite)
- docs/series2/doc_series2_fluid_bells.txt: Protocol Beta [AG1] (dual-pyramid basis)
- analysis/demos/acoustics_doc.py: AC1-AC15 (χ), SP1-SP7c (spinning+saturation+ T_{2g} +void+scaling+configs+dumbbell)
- analysis/demos/gyroscope_doc.py: GY8 (Ω^2 scaling law)
- experiments/bench_pyramid_escalator.txt: full experimental protocol
- docs/doc_series2_synthesis.txt: Section 1.2 (vortex retrodictions)

SECTION 8: CONCLUSIONS

A spinning pentagonal pyramid preserves C5 symmetry and creates a centrifugal

T_{1g} chi coupling effect on its spin axis. The predicted consequences

are specific, falsifiable, and testable for under \$30:

1. Apparent weight reduction on axis proportional to ω^2 [GY8]
2. Reversal on apex flip (null control) [SP1]
3. 4-sided pyramid produces weight INCREASE; 3-sided produces nothing [SP1]
4. $N=5$ carousel gives $\phi^2 = 2.618x$ more effect than $N=4$ [SP2]

The critical discriminator is the sign reversal between 5-sided (T_1g pressure reduction, $\chi = +\phi$) and 4-sided (compression, $\chi = -1$): conventional fluid dynamics predicts the same sign for both, the torsionverse predicts opposite signs.

Companion script: `analysis/demos/acoustics_doc.py` [27/27 PASS, SP1-SP7c].

Chi algebra is common to `doc_series2_piezo_escalator.txt` [AC1-AC14].